

$$29 - 40 - 0 = -11$$

Taller: Actividad Operadores.

1. Determine el valor de las siguientes expresiones

A.  $4 + 2 * 5$

$$4 + 10 = 14$$

B.  $23 * 2 / 5$

$$46 / 5 = 9.2$$

$$\begin{array}{r} 23 \\ \times 2 \\ \hline 46 \end{array}$$

$$\begin{array}{r} 46 \\ 10 \\ \hline 0 \end{array} \quad \begin{array}{r} 15 \\ 9.2 \end{array}$$

C.  $6 + 2 * 3 - 4 / 2$

$$6 + 6 - 4 / 2$$

$$6 + 6 - 2 \rightarrow 12 - 2 = 10$$

D.  $5 * 5 (5 + (6 - 2) + 1)$

$$5 * 5 (5 + 4 + 1)$$

$$5 * 5 = 10$$

$$25 * 10$$

$$250$$

$$\begin{array}{r} 25 \\ \times 10 \\ \hline 00 \\ 25 \\ \hline 250 \end{array}$$



$$6. 7 - 6 / 3 + 2 * 3 / 2 - 4 / 2 * 2$$

$$7 - 2 + 2 * 3 / 2 - 4 / 2 * 2$$

$$7 - 2 + 6 / 2 - 4 / 2 * 2$$

$$7 - 2 + 3 - 2 * 2$$

$$7 - 2 + 3 - 4$$

$$5 + 3$$

$$8 - 4 = 4 \quad \checkmark$$

$$F. 5 \% 2$$

↓

1

5

2

2

Residuo

$$6. 45 / 5 \% 2$$

$$9 \% 2 = 1$$

$$\begin{array}{r} 45 \\ 5 \overline{) 45} \\ 0 \end{array}$$

$$H. 3 + 5 * (10 - (2 + 4))$$

$$3 + 5 * (10 - 6)$$

$$3 + 5 * 4$$

$$3 + 20 = 23$$

$$I. 3 + 6 * (-10 - 6)$$

$$3 + 6 * -16$$

$$3 + (-96)$$

$$3 - 96 = -93$$

$$\begin{array}{r} 3 \overline{) 16} \\ 15 \end{array}$$

$$\begin{array}{r} 1 \overline{) 16} \\ 15 \end{array}$$



$$5. 3.5 + 5.09 - 14.0 / 40$$

$$8.59 - - 0.35$$

$$8.24$$

$$3.50$$

$$5.09$$

$$8.59$$

$$14.0$$

$$20$$

$$00$$

$$140$$

$$0.35$$

$$k. 3/5 + 5.09 - 3.5 + 8.59 - 3.5$$

$$3/5 + 5.09 - 3.5 + 8.59 - 3/5$$

$$5.09 - 3.5 + 8.59$$

$$1.59 + 8.59 = 10.18$$

$$159$$

$$8.59$$

$$10.18$$

$$5.09$$

$$3.50$$

$$1.59$$

$$l. 2.1 * (1.5 + 3.0 * 4.1)$$

$$2.1 * 1.5 + 12.3$$

$$2.1 * (1.5 + 12.3)$$

$$2.1 * (13.8)$$

$$28.98$$

$$3.0$$

$$x 4.1$$

$$30$$

$$120$$

$$1230$$

$$1$$

$$31.5$$

$$1.5$$

$$12.3$$

$$45.3$$

$$138$$

$$21$$

$$138$$

$$216$$

$$28.98$$

$$m. 2.1 * (1.5 - 12.3)$$

$$2.1 * -10.8$$

$$-22.68$$

$$1.5$$

$$-12.3$$

$$-10.8$$

$$1$$

$$10.8$$

$$x 2.1$$

$$20.8$$

$$21.6$$

$$22.68$$



$$1/6 \quad 15 - 3\% (4 - 2)) + 3 * 2$$

$$15 - 3\% (4 - 2)) + 6$$

$$(15 - 3\% + 2) + 6$$

$$5 + 1 + 6$$

$$4 + 6 = 10$$

$$0. 7 * 4 - 2 + 5 - (2 + 2 * 5) * 2 - 3$$

$$7 * 4 - 2 + 5 - (2 + 10) * 2 - 3$$

$$7 * 4 - 2 + 5 - (12) * 2 - 3$$

$$28 - 2 + 5 - 24 - 3$$

$$26 + 5 - 24 - 3$$

$$7 - 3 = 4$$

$$P. 8 + 7 * 3 + 4 * 6$$

$$8 + 21 + 24$$

$$29 + 24$$

$$53$$

$$Q. -2 \wedge 3$$

$$-8$$

$$-2 * -2 * -2 = 8$$

$$+ * 2 = 4$$



$$2. (33 + 3 \times 1) / 5$$

$$(33 + 12) / 5$$

$$45 / 5$$

$$9$$

$$\begin{array}{r} 45 \overline{) 5} \\ 0 \end{array}$$

$$5. 2^2 \times 3$$

$$2^2 \times 2 = 4 \times 3$$

$$12$$

$$7. 3 + 2 \times (18 - 4^2)$$

$$3 + 2 \times (18 - 16)$$

$$3 + 2 \times 2$$

$$3 + 4$$

$$7$$

$$11. 16 \times 16 - 3 \times 2$$

$$256 - 6$$

$$250$$

$$\begin{array}{r} 3 \\ \times 16 \\ 15 \\ \times 96 \\ 16 \\ 256 \end{array}$$

$$12. 5 / 2 + 20 \bmod 6$$

$$2,5 + 2$$

$$4,5$$

$$\begin{array}{r} 5 \overline{) 12} \\ 10 \end{array}$$

$$\begin{array}{r} 20 \overline{) 16} \\ 2 \end{array}$$



$$w. 4 \times 6 / 2 - 12 / 2$$

$$24 / 2 - 7,5$$

$$12 - 7,5$$

$$4,5$$

$$\begin{array}{r} 15 \\ 10 \\ 0 \end{array} \begin{array}{r} 12 \\ 7,5 \end{array}$$

$$\begin{array}{r} 24 \\ 04 \\ 0 \end{array} \begin{array}{r} 2 \\ 12 \end{array}$$

$$x. 5 \times 15 / 2 / (4 - 2)$$

$$5 \times 15 / 2 / 2$$

$$75 / 2 / 2$$

$$37,5 / 2$$

$$18,75$$

$$\begin{array}{r} 15 \\ 15 \\ 75 \end{array}$$

$$\begin{array}{r} 75 \\ 15 \\ 10 \\ 0 \end{array} \begin{array}{r} 12 \\ 37,5 \end{array}$$

$$\begin{array}{r} 37,5 \\ 17 \\ 15 \\ 10 \\ 0 \end{array}$$

$$\begin{array}{r} 12 \\ 78,75 \end{array}$$

$$y. 10 + 5 \times (2^4 3)$$

$$2 \times 2 \times 2 = 8$$

$$10 + 5 \times 8$$

$$10 + 40$$

$$50$$

$$z. 100 / 4 / 5 \times 2$$

$$100 / 4 \times 2$$

$$25 / 5 \times 2$$

$$5 \times 2$$

$$10$$

$$\begin{array}{r} 100 \\ 20 \\ 0 \end{array} \begin{array}{r} 14 \\ 25 \end{array}$$



$$aa. 15 \% 4 + 2 * 3$$

$$15 \% 4 + 6$$

$$3 + 6$$

$$9$$

$$\begin{array}{r} 15 \\ 3 \end{array} \quad \begin{array}{r} 14 \\ 3 \end{array}$$

$$bb. 25 + (10 - 2^2) * 4 / 2$$

$$25 + 6 * 4 / 2$$

$$25 + 12$$

$$37$$

$$cc. (8 + 2) * (5 - 3)^2 / 10$$

$$10 * 2^2 / 10$$

$$10 * 4 / 10$$

$$40 / 10$$

$$4$$

$$\begin{array}{r} 40 \\ 0 \end{array} \quad \begin{array}{r} 10 \\ 4 \end{array}$$



$$A = 20 \quad b = 10 \quad C = 5$$

$$a + b * 3 > 5$$

$$20 + 30 > 5$$

$$50 > 5 \text{ verdadero}$$

$$(b * 5) + a * 3/2 < b * C$$

( $\leftarrow$ ) asignar

$$50 + 60/2 < 50$$

$$X \leftarrow 3 + 5$$

$$50 + 30 < 50$$

$$X \leftarrow 8$$

$$80 < 30 \text{ Falso}$$

$$a = 5 \quad b = 10 \quad C = 20$$

$$X \leftarrow 3 + 5$$

$$X \leftarrow 8$$

Parte #2 fallar Falso o Verdadero  $A = 10 \quad B = 20 \quad C = 30$

$$a. a + b > C$$

$$X = 15 \quad Y = 10 \quad Z = 5$$

$$10 + 20 > 30$$

$$30 > 30 \text{ Falso}$$

$$b. a - b < C$$

$$10 - 20 < 30$$

$$-10 < 30 \text{ Verdadero}$$



$$C. a - b = C$$

$$10 - 20 = 30$$

$$10 = 30 \text{ falso}$$

$$D. a * b \neq C$$

$$10 * 20 \neq 30$$

$$200 \neq 30 \text{ verdadero}$$

$$E. a * b - b * a < b * 5$$

$$10 * 30 - 20 * 10 < 20 * 5$$

$$300 - 200 < 100$$

$$100 < 100 \text{ falso}$$

$$F. c * a - b < a + b + c$$

$$30 * 10 - 20 < 10 + 20 + 30$$

$$300 - 20 < 60$$

$$280 < 60 \text{ falso}$$

$$G. b + (2 - a) / 5 < a * c * 2$$

$$20 + (2 - 10) / 5 < 10 * 30 * 2$$

$$20 + -8 / 5 < 300 * 2$$

$$-12 / 5 < 600$$

$$-2,4 < 600 \text{ Verdadero}$$



$$H. 8 = 16 \vee 7! = 4 \vee 4 < 1$$

$$8 = 16 \vee 7! = 4 \vee 4 < 10$$

$$8 = 16 \text{ falso}$$

$$7! = 4 \text{ verdadero}$$

$$4 < 1 = \text{Falso}$$

falso

$$y = V + V = \text{Verdadero}$$

$$V + F = \text{Falso}$$

$$F + F = \text{Falso}$$

$$O = V + V = \text{Verdadero}$$

$$V + F = \text{Verdadero}$$

$$F + F = \text{Falso}$$

$$i. (4 * 3 < 6 \vee 3 > 5 - 2) \vee 3 + 2 < 12$$

$$4 * 3 < 6 \vee 3 > 5 - 2 \vee 3 + 2 < 12$$

$$12 < 6 \vee 3 > 3 \vee 5 < 12$$

Falso

Falso

Verdadero

falso

$$j. (x > y) \vee (z < y)$$

$$(15 > 10) \vee (5 < 10)$$

= Verdadero

verdadero

Verdadero

$$k. (x + y = 25) \vee (z > x)$$

$$(15 + 10 = 25) \vee (5 > 10) = \text{Verdadero}$$

Verdadero

falso



$$l. ! (x \neq y + z)$$

$$! (15 \neq 10 + 5)$$

$$\begin{array}{r} 15 \quad | \quad 2 \\ 1 \quad | \quad 7 \end{array}$$

$$! (15 \neq 15) = \text{Verdadero}$$

$$m. (x \% z == 0) \text{ o } (y \% z == 0)$$

$$(15 \% 2 == 0) \text{ o } (10 \% 2 == 0)$$

$$(1 == 0) \text{ o } (0 == 0) = \text{Verdadero}$$

Falso                      Verdadero

$$n. (x - z >= y) \text{ y } (z * 2 == y)$$

$$(15 - 5 >= 10) \text{ y } (5 * 2 == 10) \quad \left. \vphantom{(15 - 5 >= 10)} \right\} \text{verdadero}$$

$$10 >= 10 \text{ y } (10 == 10)$$

3. Deducir el valor  $A=5$   $B=25$   $C=10$

$$x \leftarrow A + B + C$$

$$x \leftarrow 5 + 25 + 10$$

$$x \leftarrow 40$$

$$x \leftarrow A + B * C$$

$$x \leftarrow 5 + 25 * 10$$

$$x \leftarrow 5 + 250$$

$$x \leftarrow 255$$



$$x \leftarrow A + B / C$$

$$x \leftarrow 5 + 25 / 10$$

$$x \leftarrow 5 + 2,5$$

$$x \leftarrow 7,5$$

$$x \leftarrow A + B \setminus C$$

$$x \leftarrow 5 + 25 \setminus 10$$

$$x \leftarrow 5 + 2,5$$

$$x \leftarrow 7,5$$

$$x \leftarrow A + B \bmod C$$

$$x \leftarrow 5 + 25 \bmod 10$$

$$x \leftarrow 5 + 5 + 10$$

$$x \leftarrow 20$$

$$x \leftarrow (A + B) \setminus C$$

$$x \leftarrow (5 + 25) / 10$$

$$x \leftarrow 30 / 10$$

$$x \leftarrow 3$$



$$x \leftarrow A + (B / C)$$

$$x \leftarrow 5 + (25 / 10)$$

$$x \leftarrow 5 + 2,5$$

$$x \leftarrow 7,5$$

4. Que se obtiene en las variables A y B, despues de la ejecución de las siguientes instrucciones

$$A = 5$$

$$A = \underline{5}$$

$$B = A + 6$$

$$B = 5 + 6 = \underline{11}$$

$$A = A + 1$$

$$A = 5 + 1 = \underline{6}$$

$$B = A - 5$$

$$B = 6 - 5 = \underline{1}$$

$$A = \underline{6}$$

$$B = \underline{1}$$

5. Encuentre el valor final de las variables A y B

$$A = 3$$

$$A = \underline{3}$$

$$B = 5$$

$$B = \underline{5}$$

$$A = B + 3$$

$$A = 5 + 3 = \underline{8}$$

$$B = A * 2$$

$$B = 8 * 2 = \underline{16}$$

$$B = B - 2$$

$$B = 16 - 2 = \underline{14}$$

$$A = B * 3$$

$$A = 14 * 3 = \underline{42}$$

$$A = A + B$$

$$A = 42 + 14 = \underline{46}$$

$$A = \underline{46}$$

$$B = \underline{14}$$



## 6. Seguir el algoritmo (Traza)

## a. Ejercicio A:

1.  $P \leftarrow 10$

$P \leftarrow \underline{10}$

2.  $Q \leftarrow 20$

$Q \leftarrow 20$

3.  $P \leftarrow P + Q$

$P \leftarrow 10 + 20 = \underline{30}$

4.  $Q \leftarrow P - Q$

$Q \leftarrow 30 - 20 = \underline{10}$

5.  $P \leftarrow P - Q$

$P \leftarrow 30 - 10 = \underline{20}$

$P = \underline{20}$

$Q = \underline{10}$

## Ejercicio B

1.  $x \leftarrow 2$

$x \leftarrow \underline{2}$

2.  $y \leftarrow 3$

$y \leftarrow \underline{3}$

3.  $z \leftarrow x \wedge y$

$z \leftarrow 2 \wedge 3 = \underline{0}$

4.  $x \leftarrow z * y$

$x \leftarrow 0 * 3 = \underline{0}$

5.  $y \leftarrow x \% z$

$y \leftarrow 0 \% 0 = \underline{0}$

$x = \underline{0}$

$y = \underline{0}$

$z = \underline{0}$



7. Deducir el valor  $A = 8$   $B = 3$   $C = -5$

a.  $a + b + c$

$$8 + 3 + (-5)$$

$$11 - 5 = 6$$

b.  $2 * b + 3 * (a - c)$

$$2 * 3 + 3 * (8 - (-5))$$

$$2 * 3 + 3 * (13)$$

$$6 + 39$$

$$45$$

c.  $a / b$

$$8 / 3$$

$$\begin{array}{r} 8 \overline{) 24} \\ 20 \end{array} \begin{array}{r} 13 \\ 2,6 \end{array}$$

$$2,6$$

d.  $a \bmod b$

$$8 \bmod 3$$

$$2$$

e.  $a / c$

$$8 / -5 = -1,6$$

$$\begin{array}{r} 8 \overline{) -8} \\ 30 \end{array} \begin{array}{r} -5 \\ 1,6 \end{array}$$



$$E. a \bmod c$$

$$8 \bmod -5$$

$$3$$

$$G. a * b / c$$

$$8 * 3 / 5$$

$$24 / 5$$

$$4,8$$

$$\begin{array}{r} 24 \overline{) 15} \\ 10 \phantom{0} \\ \hline 4,8 \\ 0 \end{array}$$

$$H. a * (b / c)$$

$$8 * (3 / -5)$$

$$24 / -5 = 4,8$$

$$\begin{array}{r} 24 \overline{) -15} \\ 40 \phantom{0} \\ \hline 4,8 \\ 0 \end{array}$$

$$I. (a * c) \bmod b$$

$$(8 * (-5)) \bmod 3$$

$$-40 \bmod 3$$

$$1$$

$$\begin{array}{r} -40 \overline{) 13} \\ 10 \phantom{0} \\ \hline -13,3 \\ 10 \phantom{0} \\ \hline 1 \end{array}$$

$$J. a * (c \bmod b)$$

$$8 * (-5 \bmod 3)$$

$$8 * -2$$

$$-16$$

$$\begin{array}{r} -5 \overline{) 13} \\ 20 \phantom{0} \\ \hline -13,6 \\ -2 \end{array}$$



8. Calcular el valor de la variable

a. Valor  $\leftarrow 4.0 + 5$

Valor  $\leftarrow 20.$

b.  $x \leftarrow 3.0$

$y \leftarrow 2.0$

Valor  $\leftarrow x^y - y$

Valor  $\leftarrow 3.0^{2.0} - 2.0$

Valor  $\leftarrow 9.0 - 2.0$

Valor  $\leftarrow 7.0$

c. Valor  $\leftarrow 5$

$x \leftarrow 3$

Valor  $\leftarrow \text{Valor} * x$

Valor  $\leftarrow 5 * 3$

Valor  $\leftarrow 15$

9. Deducir valor de A y B

a.  $a \leftarrow 4$

$a \leftarrow \underline{4}$

$b \leftarrow a$

$b \leftarrow \underline{4}$

$b \leftarrow a + 3$

$b \leftarrow 4 + 3 = \underline{7}$

$A = 4$

$B = 7$



$$b. \quad a \leftarrow 5$$

$$b \leftarrow a + 6$$

$$a \leftarrow a + 1$$

$$b \leftarrow a - 5$$

$$a \leftarrow \underline{5}$$

$$b \leftarrow 5 + 6 = \underline{11}$$

$$a \leftarrow 5 + 1 = \underline{6}$$

$$b \leftarrow 6 - 5 = \underline{1}$$

$$A = \underline{6}$$

$$B = \underline{1}$$

$$C. \quad a \leftarrow 3$$

$$b \leftarrow 20$$

$$c \leftarrow a + b$$

$$b \leftarrow a * b$$

$$a \leftarrow b - c$$

$$a \leftarrow \underline{3}$$

$$b \leftarrow \underline{20}$$

$$c \leftarrow 3 + 20 = \underline{23}$$

$$b \leftarrow 3 * 20 = \underline{23}$$

$$a \leftarrow 23 - 23 = \underline{0}$$

$$A = \underline{0}$$

$$B = \underline{23}$$

$$C = \underline{23}$$

$$D. \quad a \leftarrow 10$$

$$b \leftarrow 5$$

$$a \leftarrow b$$

$$b \leftarrow a$$

$$a \leftarrow \underline{10}$$

$$b \leftarrow \underline{5}$$

$$a \leftarrow \underline{5}$$

$$b \leftarrow \underline{10}$$

$$A = \underline{5}$$

$$B = \underline{10}$$